ME 2160 Introduction to Mechanical Engineering Design Final Project

Design, Build, Test, and Analysis of an Electrically Powered Vehicle

Friday December 5th, 2025

Ethan Jiang, Wonsun You

Abstract:

This report describes the design, development, and testing of the ‘Blue’ car, an electrically powered vehicle inspired by the form of a blue whale. The project required teams to construct a motor driven vehicle using laser-cut 1/8" plywood components and kit-supplied mechanical elements like gears. The primary performance objective was to travel at least 10 feet unaided, with additional incentives for speed, compactness, aesthetics, and creativity.

Blue incorporates a gear transmission driven by the kit DC motor, using an 11-tooth pinion engaging a 45-tooth output gear to achieve a 4.1:1 gear ratio. The chassis and structural supports were laser cut from plywood, while kit wheels and metal rod axles provided reliable rolling performance. The final mass of the vehicle was 162.8 g. Aesthetic elements include a whale face, side flippers, a rear tail fin, and a top-mounted blowhole switch, chosen to create a recognizable marine-themed design while maintaining mechanical function.

Testing demonstrated that Blue successfully traversed the required 10 feet in approximately 1.8 seconds, with a slight drift to the right attributed to minor mismatches in the chassis and rod. Earlier prototypes suffered from gear grinding and structural looseness, but iterative improvements to gear selection and alignment significantly enhanced performance. The project provided experience in mechanical design, transmission tuning, structural layout, and integration of aesthetic and functional elements within a cohesive engineered system.

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Introduction:

The ME2160 team project required students to design, build, and test an electrically powered vehicle capable of traveling at least 10 feet unaided. The vehicle was required to incorporate gears, belts, or pulleys for power transmission, rely on laser-cut 1/8" plywood for the primary structure, and fit within an 8" × 8" × 8" volume. Additional points were awarded for categories such as fastest, lightest, most compact, most school spirit, and highest "coolness factor."

Our team is Blue, a vehicle themed after a blue whale. The project aimed to balance mechanical functionality with distinctive aesthetic/marine design. Blue features a whale face on the front, side flippers, a tail fin, and a blowhole switch integrated into the electrical system. Mechanically, the vehicle incorporates a gear transmission selected for both speed and reliability.

This report presents the full design process for Blue, including background research, functional goals, task specifications, design development, experimental results, and reflections.

Background Research:

The project's initial phase focused on understanding the requirements and constraints associated with small electric vehicles and compact transmission systems. Research centered on three core areas: drivetrain behavior, gear ratio selection, and thematic body integration.

Small vehicle performance is heavily influenced by rolling resistance, friction at axle supports, wheel alignment, and total mass. Minimizing friction and maintaining straightness were identified as key factors for achieving fast and consistent travel. The provided metal rod axles and kit wheels offered low rolling resistance when properly aligned.

Gear ratio selection required balancing rotational speed and output torque. A higher reduction ratio increases torque while lowering wheel speed; conversely, a lower ratio increases speed but risks insufficient torque for startup motion. The assignment's recommended minimum ratio of 3:1 guided our exploration of gear combinations. Initial testing showed that overly aggressive ratios with large mismatches between gear sizes caused grinding and inefficient transmission. A more moderate ratio eliminated these issues.

Aesthetic research focused on translating recognizable features of blue whales into structural forms compatible with the vehicle's function. The body design included a rounded head, elongated shape, side flippers, a rear tail fluke, and a blowhole. These elements were designed to enhance appearance without significantly increasing mass or interfering with mechanical components.

The constraints on vehicle size and mandatory use of laser-cut plywood ensured that the structural and aesthetic choices aligned with the project's fabrication capabilities.

Goal Statement:

The goal of this project is to design, build, and test an electrically powered vehicle capable of reliably traversing 10 feet unaided at high speed while maintaining stability and clearly embodying a blue whale theme. Functionally, the vehicle must convert electrical power into rotational motion through a gear transmission, transmit this motion effectively to the wheels, remain structurally stable, and operate without tipping or component detachment.

The vehicle must also communicate its whale theme through recognizable visual features including a whale face, flippers, tail fin, and blowhole.

Task Specifications:

The project requirements specified that the vehicle must be electrically powered using the motor(s) supplied in the kit and incorporate a transmission that uses gears, belts, and/or pulleys. All structural components were required to be fabricated from laser-cut 1/8-inch plywood, and the completed vehicle had to fit within an 8" × 8" × 8" volume. To receive full performance credit, the vehicle needed to traverse at least 10 feet unaided and be painted or decorated to highlight team identity or theme. Additional points were available for achieving distinctions such as lightest weight, most school spirit, most compact design, fastest speed, highest "coolness factor," most Rube Goldberg–like design, slowest motion, or greatest towing capacity. For performance validation, the vehicle was required to demonstrate autonomous motion over the test distance, operate without external assistance once activated, retain all components during operation, and comply with all size, material, and fabrication constraints established in the assignment.

Design Description:

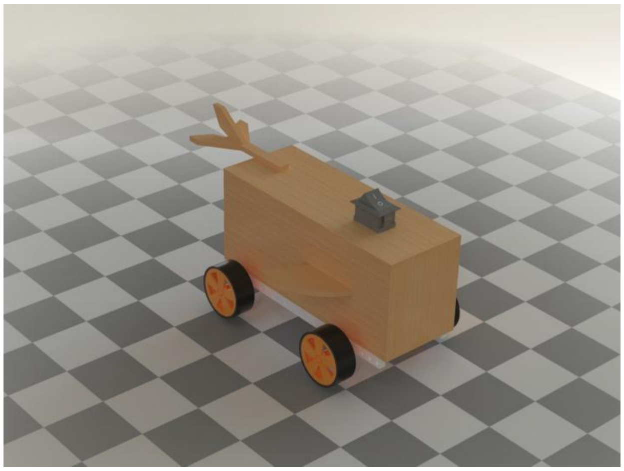

Figure 1: Front and side view of car. (Rendered picture)

Our car was designed as a compact four-wheeled vehicle incorporating a whale inspired body. The side flippers extend laterally to reinforce the theme, and the tail fin occupies the rear. The blowhole, positioned on the top of the body, acts as the electrical switch. The aesthetic elements are layered over a plywood chassis that supports the drivetrain and electrical system.

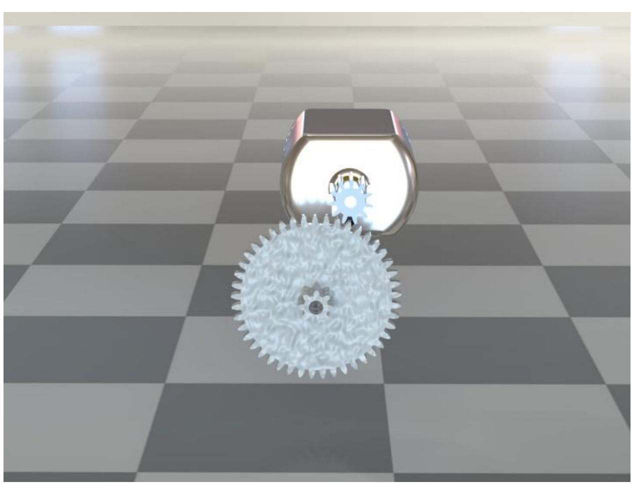

Figure 2: Isolated 1:1 CAD model of gear transmission with a gear ratio of 4.1:1

The vehicle uses the provided DC motor from the hardware kit. The motor drives an 11-tooth gear, which meshes with a 45-tooth output gear, yielding a ratio of approximately 4.1:1. This ratio was selected after multiple iterations involving gears with larger mismatches, which caused excessive grinding and poor meshing. The output shaft is rigidly coupled to the rear wheels, allowing both wheels to contribute to propulsion. The front axle is free-rolling to reduce friction and maintain directional stability.

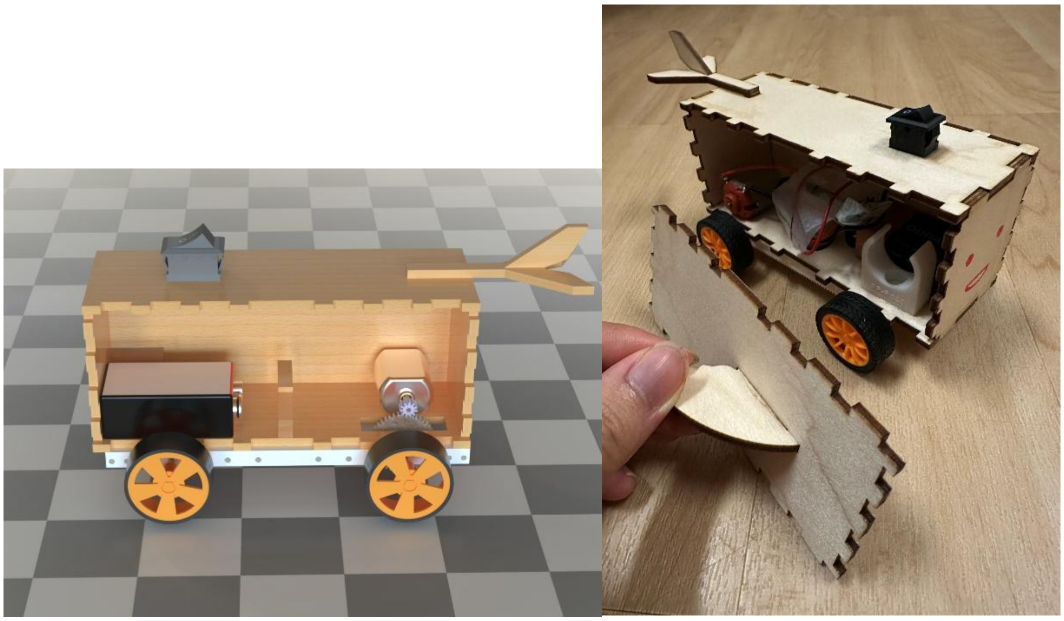

Figure 3: Side and inside view of the car

The chassis is constructed from laser-cut 1/8" plywood, forming a rectangular base with side supports for the drivetrain and body attachment. Metal rod axles pass through holes in the plastic chassis supports, providing stiffness and resistance to deformation. The battery pack and motor are mounted low within the frame and on the edgers of the car to maintain a stable center of mass. Kit wheels are press-fit onto the axles to ensure consistent rotation. The side pod was not glued on and instead fit in like a puzzle for inside access of the car. Although making this more accessible, this panel experienced loosening under dynamic loads during testing.

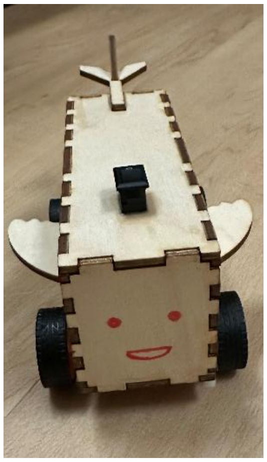

Figure 4: Front and detailed view displaying the whale

The whale body consists of laser-cut side panels and top panels attached to the chassis. The front face includes painted details to represent eyes and mouth. Side flippers extend from the lower chassis area, and the tail fin occupies the rear. The blowhole switch is integrated structurally and electrically as a functional control element. A side pod was added to allow access to the vehicle's interior components for wiring adjustments and battery replacement. Also, axle supports were checked for perpendicularity to prevent binding and uneven rolling resistance. The whale body panels were attached after drivetrain testing to avoid interference with mechanical components.

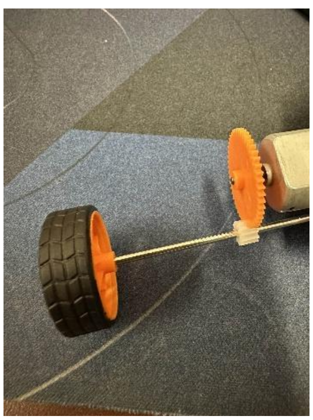

Figure 5: Photo of a previous iteration transmission with a gear ratio of 1:4.1

Early prototypes used gear combinations with extreme size differences, resulting in grinding, vibration, and inconsistent motion. Adjustments to gear spacing and shaft alignment improved performance but did not fully eliminate mechanical issues.

The final gear ratio of 4.1:1 produced smooth motion and significantly increased speed. Wheel alignment adjustments also improved directional stability, though a slight drift to the right remained. Decorative components were progressively refined to avoid interfering with the wheels and drivetrain.

The finished CAD assembly produced a mass of 159.49 grams as shown below.

```
Report coordinate values relative to: -- default --

Mass properties of car
    Configuration: Default
    Coordinate system: -- default --

Mass = 159.49 grams

Volume = 137644.12 cubic millimeters

Surface area = 109549.33  square millimeters

Center of mass: ( millimeters )
     X = 571.52
     Y = 927.37
     Z = 1345.89

Principal axes of inertia and principal moments of inertia: ( grams *  square millim
Taken at the center of mass.
      Ix = ( 0.99,  0.03, -0.10)       Px = 57260.26
      Iy = (-0.10, -0.19, -0.98)       Py = 255569.88
      Iz = (-0.05,  0.98, -0.18)       Pz = 262777.58

Moments of inertia: ( grams *  square millimeters )
Taken at the center of mass and aligned with the output coordinate system. (Usin
     Lxx = 59547.53              Lxy = 5998.80               Lxz = -20389.10
     Lyx = 5998.80               Lyy = 262351.47             Lyz = 720.70
     Lzx = -20389.10             Lzy = 720.70                Lzz = 253708.71

Moments of inertia: ( grams *  square millimeters )
Taken at the output coordinate system. (Using positive tensor notation.)
     Ixx = 426121648.25          Ixy = 84536198.96           Ixz = 122657235.
     Iyx = 84536198.96           Iyy = 341254668.48          Iyz = 199064383.
     Izx = 122657235.57          Izy = 199064383.50          Izz = 189510906.
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Figure 6: Mass properties of the ‘Blue’ car

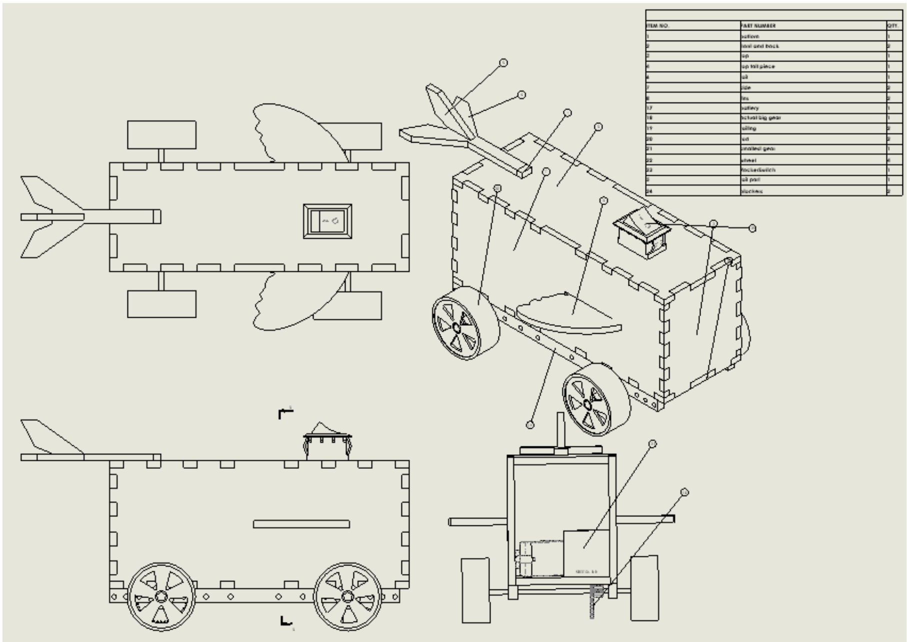

Figure 7: Assembly Drawing with Bill of Materials Table

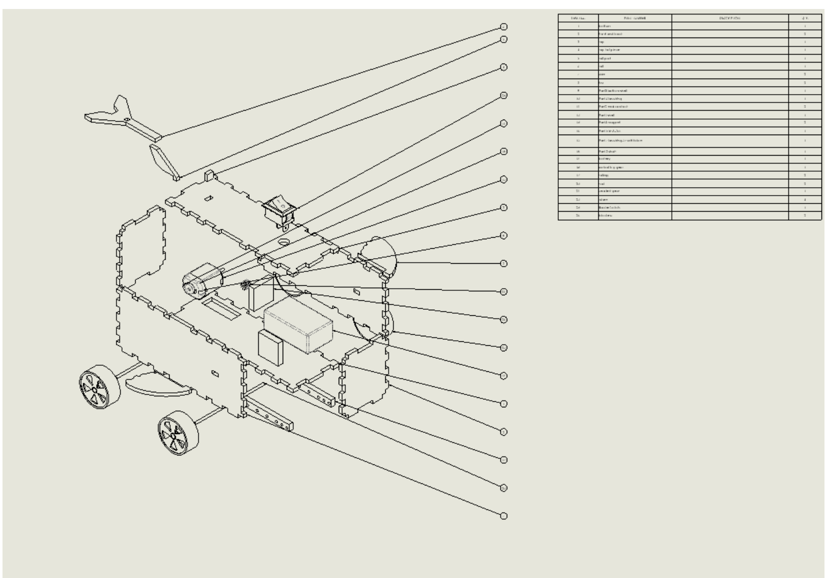

Figure 8: Exploded Assembly Drawing with Bill of Materials Table

Results:

The car successfully traversed the required 10-foot distance with a measured time of 1.8 seconds. This performance meets the project requirement and is competitive among vehicles emphasizing speed. The vehicle exhibited a slight drift to the right, likely due to an error in glueing the chassis to the rod as shown below.

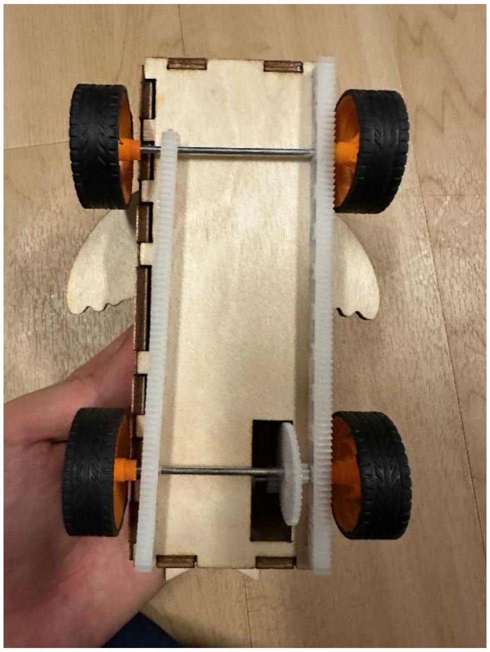

Figure 9: Photo of the chassis from below. The left chassis is crooked to the right.

Despite this drift, the vehicle remained stable and did not tip during operation.

The drivetrain functioned reliably in the final configuration, with no recurrence of the gear grinding observed in earlier prototypes. The primary structural issue encountered was the loosening of the side pod, which nearly detached during testing. All other body elements remained secure and did not adversely affect performance.

Aesthetically, the vehicle clearly resembled a blue whale, fulfilling the thematic objective and contributing positively to the project's creative aspect.

Conclusion:

The development of the 'Blue' car demonstrated the value of iterative engineering design, particularly in the areas of drivetrain tuning and structural integration. The project reinforced the importance of selecting appropriate gear ratios, ensuring precise alignment, and minimizing friction to maximize performance. The transition from grinding early prototypes to a smooth and fast final design highlighted the role of small geometric adjustments in mechanical systems.

The project also illustrated how aesthetic and functional components must be integrated thoughtfully. While the whale body enhanced visual appeal, the loosening side pod emphasized the need for structurally robust attachment methods, even for decorative elements.

This experience provided meaningful exposure to rapid prototyping, transmission design, and physical testing. It also emphasized the need for teamwork, communication, and clear documentation. Future improvements would include reinforcing the side pod attachment, further refining wheel alignment to eliminate drift, and exploring alternative gear ratios or wheel sizes to approach the aspirational 1-second target for the 10-ft run.

Bibliography:

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Appendices:

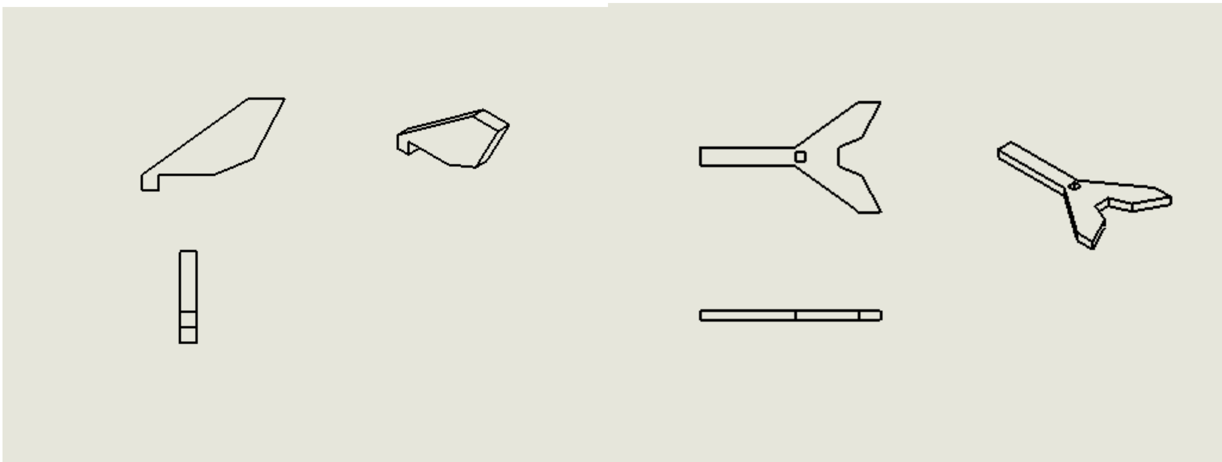

Figure 10: Drawing of the tail section of the vehicle shell

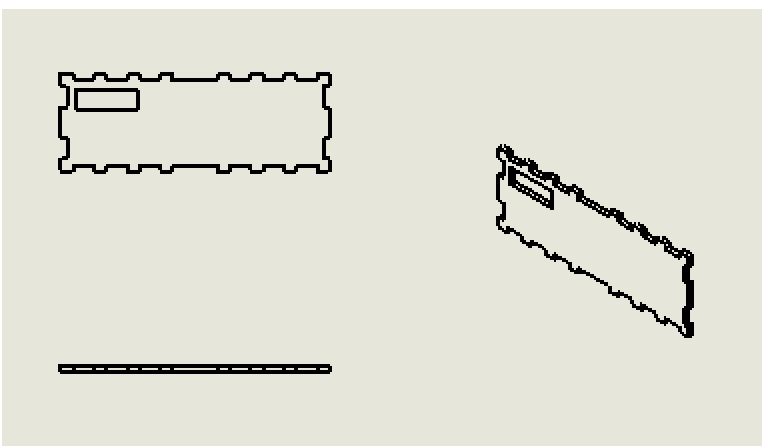

Figure 11: Drawing of the bottom section of the vehicle shell

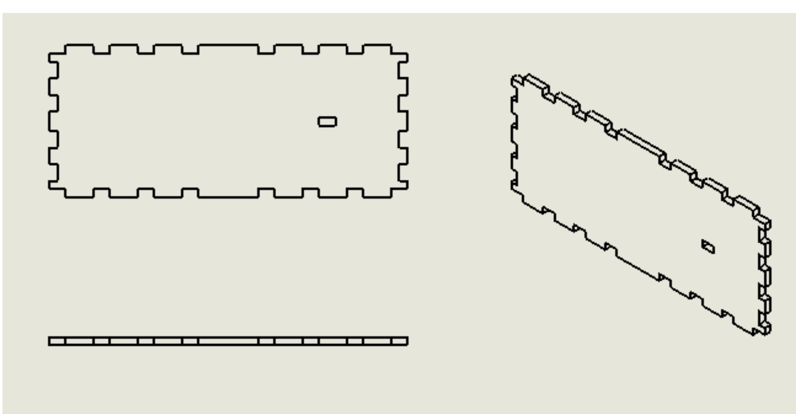

Figure 12: Drawing of the side section of the vehicle shell

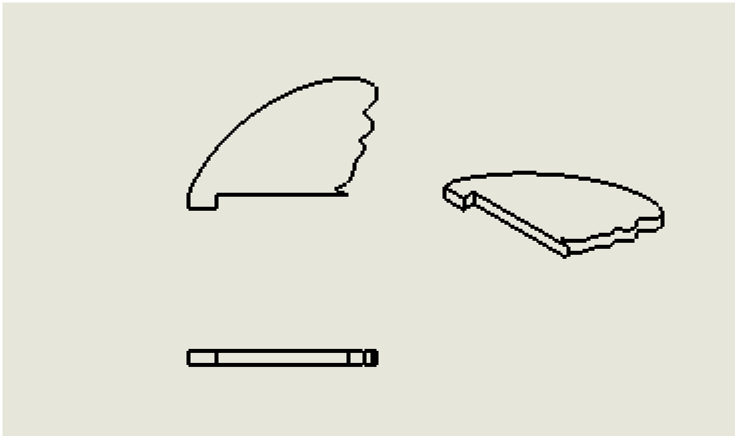

Figure 13: Drawing of the flap section of the vehicle shell

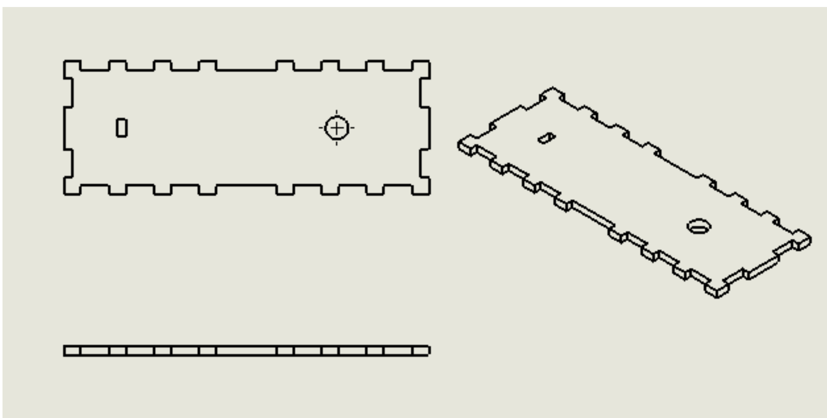

Figure 14: Drawing of the top section of the vehicle shell

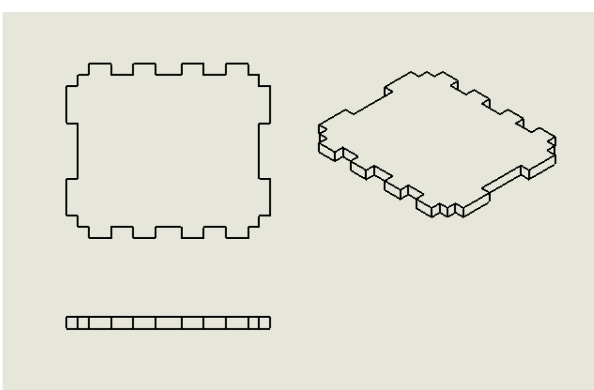

Figure 15: Drawing of the front and rear section of the vehicle shell